

In-Database Machine Learning: A Study on SQL-Based Predictions

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Abstract—The inference of machine learning models in enterprise systems often involves the transfer of information outside of the Database Management System (DBMS) and therefore incurs latency and complexity issues. This drawback is an incentive to incorporate predictive intelligence into database systems to enable effective real-time analytics without using any data processing. This paper proposes a framework for intelligent in-database machine learning, where prediction models are directly derived from SQL queries through the translation of decision trees into SQL CASE statements via middleware-based rewriting of SQL queries. It also proposes a cost-based optimizer, which it calls the Cost-Based Optimizer (CBO), and a role-based access control component, which it calls the Role-Based Access Control (RBAC), for the efficient management of the in-database machine learning models. Additionally, there is a cost-based reinforcement learning component based on the Double Deep Q-Network (Double DQN) method with Prioritized Experience Replay (PER). The architecture integrates both supervised models of learning and reinforcement learning based optimization into one structure to allow the adaptive and effective decision making in database systems. The performance of the framework is also compared with existing approaches on predictive maintenance, banking marketing, and telecom churn datasets and shows competitive performance without incurring inference latency. It also includes Explainability of AI through the use of SHAP. Based on the relational database system, it allows providing scalable, interpretable, and cost-aware analytics.

Index Terms—In-Database Machine Learning, SQL translation, Reinforcement Learning, Cost Optimisation, Intelligent middleware.

I. INTRODUCTION

As data-driven decision-making systems have grown quickly, more and more businesses are using Machine Learning (ML) in their systems, financial analytics, industrial automation, telecommunications, and Internet of Things (IoT) applications. The majority of deployment architectures separate the Database Management System (DBMS)

and the ML inference engines, even though the ML models are very good at predictions. The usual way to use data is to take it out of the relational database, change it outside of the database, and then use it for reporting or analytics.

In-database machine learning has been developed as an efficient solution for addressing the above limitations. However, the existing in-database machine learning solution tools have limitations associated with loosely coupled architecture, model services, or user functions, which may not support adaptability, cost management, or interpretability. In addition, the availability of ML model translation into SQL logic for database execution is limited.

In order to resolve these issues, this paper has introduced the concept of an “Intelligent SQL-ML Engine” that is a framework which supports the execution of supervised learning and reinforcement learning models by means of declarative SQL queries through the support of middleware-based query rewriting techniques. In our proposed architecture, we have transformed our trained Decision Tree Classifier model into SQL CASE statements through the support of PREDICT(failure) operator. In addition, our proposed framework is equipped with Cost-Based Optimizer, Role-Based Access Control, and Monitoring. Further more, reinforcement learning module a concern based on Double Deep Q-Network with Prioritized Experience Replay is included in our proposed framework for optimization of decisions made during the maintenance process. Additionally, the proposed framework is also enabled with ‘Explainability of AI’ using SHAP-based global and instance-level explanations. The experimental results have been achieved by our proposed framework with the support of efficient In-Database Analytics.

II. RELATED WORK

In-database machine learning has been recognized as a promising technique to remove data movement overhead and support predictive inference within relational database systems [13], [15]. This helps in facilitating tasks like machinery health prognosis and data-driven monitoring for machines in industrial settings [11], [12], [14]. Makrynioti *et al.* introduced a declarative framework named SQL4ML, in which the workflow of supervised learning is completely expressed in SQL. This approach demonstrates the feasibility of native database support for machine learning models [1], [16]. Similarly, Del Buono *et al.* proposed a framework called MASQ, where machine learning models [4] are translated into SQL queries and executed directly within the database environment [2], [3].

These models prove the feasibility of the transformation of machine learning models into SQL. However, these models do not include adaptive cost optimization techniques or reinforcement learning-based decision support systems [7], [9], which can further enhance the efficiency of the execution of the machine learning models in the database systems [10].

The second significant research direction in the field of database machine learning is the integration of the learned models with the internal architecture of the database systems. Kraska *et al.* proposed the concept belong from index data structures, where the conventional data indexing techniques are replaced with the machine learning models for the efficient access of the data [5].

Reinforcement learning [8] has shown significant promise for intelligent database optimization. Marcus and Papaemmanouil applied deep reinforcement learning for join order enumeration, demonstrating that the technique can attain better performance than other heuristic-based optimizers [6].

III. PROPOSED SYSTEM

The paper presents the development of an Intelligent SQL-ML Engine that is capable of executing predictions from machine learning models within the database system itself. The overall goal of the framework is to eliminate the need to move data between database systems and external machine learning services, while at the same time allowing efficient prediction using SQL queries.

A. System Overview

The system under development is based on a layered architecture, which consists of four layers. The Database Layer is in charge of storing structured data on the operation of machines, which is used in the prediction and training process. The Machine Learning Layer has predictive models for failures using supervised learning and optimization of the maintenance process using reinforcement learning. The Middleware Layer is in charge of capturing the SQL queries and integrating the prediction in the database query pipeline. The SQL queries are run and the results are displayed by the User Interface layer.

B. Supervised Learning Models

For the purpose of forecasting machine failure conditions, several supervised learning algorithms have been put into practice. Logistic Regression, Decision Tree, Gradient Boosting, XGBoost, are some of the supervised algorithms. The sensor variables of the machine, such as temperature, speed, torque, and tool wear, have been used to train the machine learning algorithms. Because they can handle the nonlinearity of the machine working data, ensemble algorithms like Gradient Boosting and XGBoost have been found to be the most reliable for prediction based on the experimental results.

C. Reinforcement Learning Optimization

Reinforcement learning techniques were added to the framework to help make smart maintenance decisions. The algorithms that were used are Deep Q-Network, Double Deep Q-Network, and Prioritized Experience Replay DQN(PER-DQN).

D. System Workflow

The workflow starts with the user submitting the SQL query along with the parameters of the machine. The middleware adds prediction logic to the process of running queries and calls on the trained models. Then, the system makes pre-dictions and explanations, which are sent back through the interface for monitoring and helping with decisions.

IV. SYSTEM ARCHITECTURE

This section shows the architecture of the proposed framework, incorporates the supervised learning, SQL-based inference, explainability, and reinforcement learning optimization under a relational database environment.

A. Data Initialization and Data Preparation

For this purpose, experiments are carried out on structured data sets, and the data sets used are Predictive Maintenance, Bank Marketing, and Telecom Churn. These data sets contain operational variables, which are temperature, rotational speed, torque, and tool wear, and these are used to predict failures in machines.

Before the model is trained, the data preparation is done where the irrelevant attributes are removed, and the prepared data is stored in SQLite, a relational database system, which allows machine learning algorithms to operate on the data without the need to transfer the data.

B. Supervised Model Training Phase

Two supervised learning models were implemented for the predictive analysis task:

- **Decision Tree:** This model offers prediction logic based on splitting the features.
- **Gradient Boosting:** This model uses the prediction performance based on the ensemble learning of weak learners.

The Decision Tree model is translated into SQL to be inferred in the database but the Gradient Boosting model is taken as a reference point to the performance of prediction.

C. SQL-Based Prediction Integration

The Decision Tree model is translated into the SQL logic using nested CASE WHEN statements. This enables predictions to be executed directly within the SQL queries using the following operation:

```
SELECT *, PREDICT(failure) FROM machines;
```

Fig. 1 shows the proposed system architecture which includes database storage, machine learning models, and middleware based query execution.

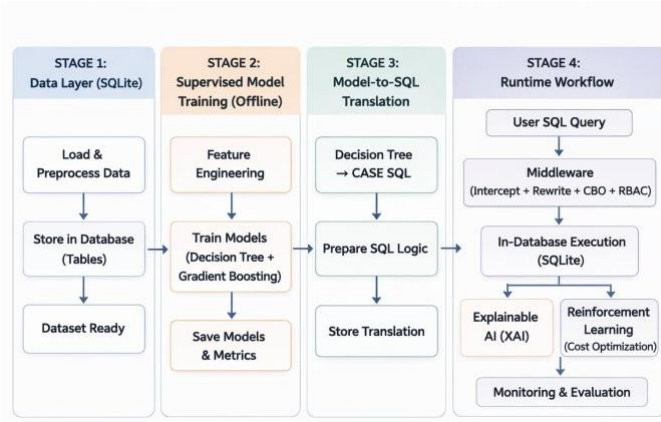


Fig. 1. Overall architecture of the proposed in-database machine learning framework.

D. Middleware Query Processing

The middleware layer receives the SQL queries of the users, and the queries are rewritten by injecting the model logic into the query pipeline. The Cost-Based Optimizer (CBO) is used to analyze the complexity of the queries which ensures efficient execution strategies.

Additionally, Role-Based Access Control (RBAC) is implemented for controlling the system, allowing administrators to monitor explanations when the user runs prediction queries.

E. Reinforcement Learning Optimization

A reinforcement learning module based on the Double Deep Q-Network method with Prioritized Experience Replay is incorporated for cost based maintenance decision support. Machine sensor variables are used to depict the state of the system, and the actions available to the system are maintenance, replacement, or operation.

Fig. 2 shows the architecture of the reinforcement learning optimization module.

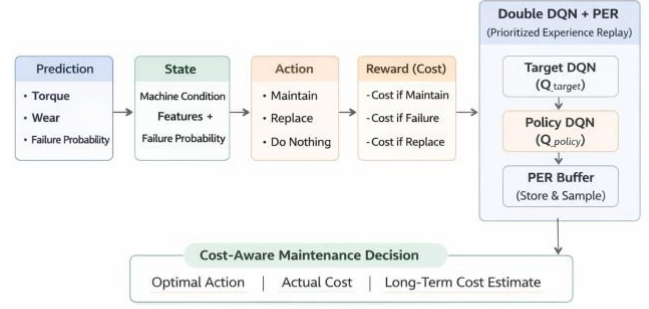


Fig. 2. Internal architecture of the reinforcement learning module using Double DQN with Prioritized Experience Replay.

V. DATASET ANALYSIS

In this study, the Predictive Maintenance dataset is employed to examine the operational behavior of machines in industrial environments and to test the efficacy of various machine learning and reinforcement learning techniques for predicting failures and optimizing the maintenance of machines. The Predictive Maintenance dataset has different types of sensor readings that measure the working conditions of the machines during the production cycles. The main types of sensor readings are air temperature, process temperature, rotation speed, torque, and tool wear.

A. Feature Correlation Analysis

Fig. 3 illustrates the correlation heatmap obtained using the dataset. The results show that air temperature and process temperature have a strong positive correlation since both measured variables relate to the temperature of the machine during operation. On the other hand, the rotation speed and torque variables show an inverse relationship resulting from changes in the mechanical load state. The tool wear variable is moderately correlated with other operational variables, indicating its effect on the progressive degradation of the machine.

B. Supervised Learning Dataset Evaluation

The dataset was also analyzed using various supervised learning algorithms to understand the behavior of machine operating features based on the interpretation of each model. The implementations of the algorithms include Logistic Regression, Decision Tree, Gradient Boosting, AdaBoost, XGBoost.

Fig. 4 demonstrates the comparative behavior of the models. From the comparison, ensemble-based models have shown better results in understanding the complex relationships between sensor variables, whereas linear decision boundaries have been observed for other models.

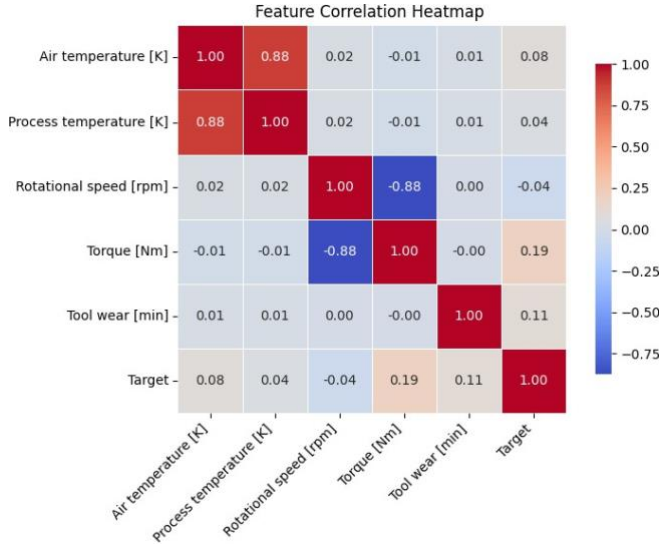


Fig. 3. Feature correlation heatmap showing relationships among machine operational variables in the predictive maintenance dataset.

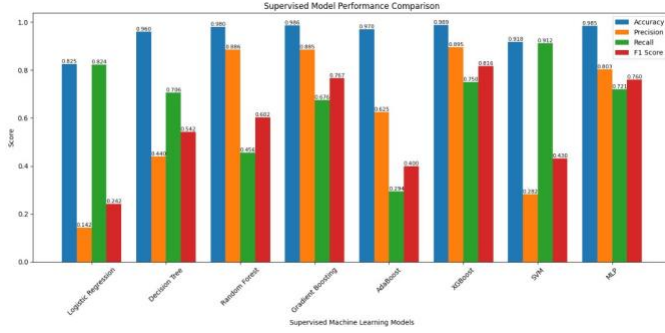


Fig. 4. Comparative behavior of supervised learning models when trained on the predictive maintenance dataset.

C. Reinforcement Learning Dataset Interaction

In order to evaluate the strategies for the maintenance decision-making process, reinforcement learning algorithms were used within the dataset environment. This is because the algorithms interact with the state variables of the machine and make policies based on the conditions of operation.

The reinforcement learning models used in this paper include the Deep Q-Network (DQN), the Double Deep Q-Network (Double-DQN), and the Prioritized Experience Re-play DQN (PER-DQN). Fig. 5 shows the learning behavior of the models and explains the learning behavior of the reinforcement learning agents.

D. Model Discrimination Analysis

The discriminatory power of the various algorithms in correctly identifying between normal and failure states was determined using the Receiver Operating Characteristic (ROC) curve.

Fig. 6 shows the ROC curves generated by all implemented models. These curves show how each algorithm reacts to

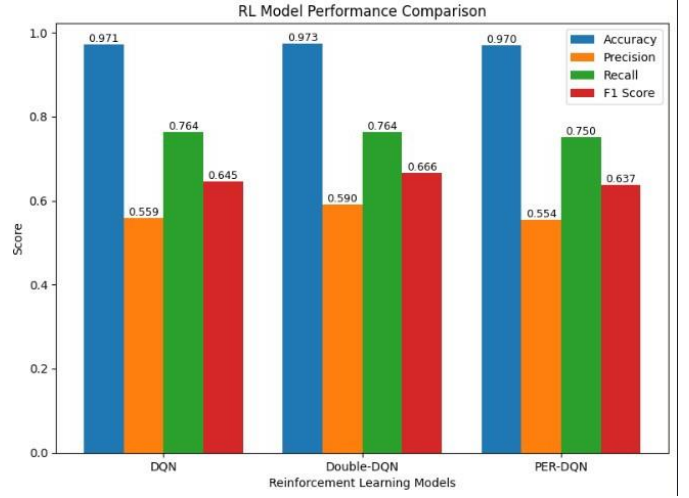


Fig. 5. Comparison of reinforcement learning algorithms interacting with machine operational states derived from the dataset.

changes in classification thresholds and demonstrate that all ensemble algorithms provide stable classification boundaries when used to classify machine operation data.

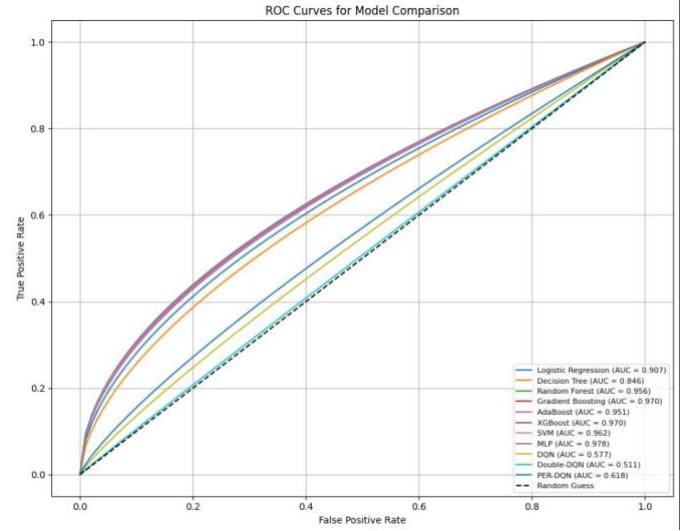


Fig. 6. ROC curve visualization illustrating classification behavior of implemented algorithms on the predictive maintenance dataset.

E. Maintenance Cost Behavior

The dataset was used to simulate the strategies of maintenance using reinforcement learning. The models derive their strategies based on the state of the operation of the machines, leading to the minimization of the overall cost of maintenance. The trend of the cost of maintenance using the reinforcement learning strategies is presented in Fig. 7. The comparison shows that the adaptive learning strategies reduce the operational cost over time.

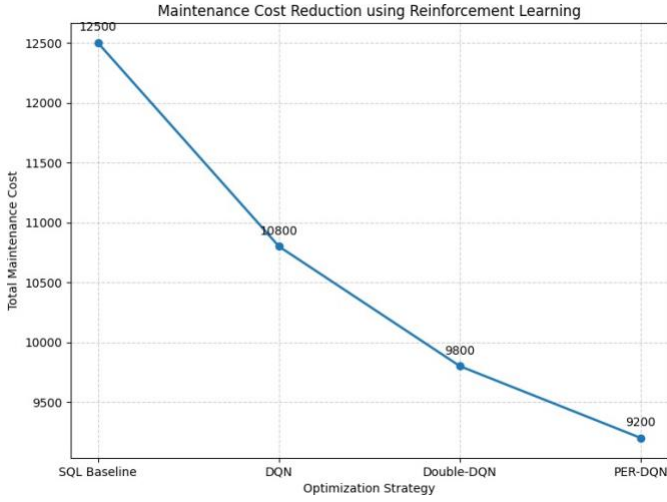


Fig. 7. Maintenance cost trend obtained through reinforcement learning based maintenance optimization.

VI. IMPLEMENTATION

In this section, the experimental setting, system configuration, training procedure, and implementation of the proposed Intelligent SQL-ML Engine framework are discussed. The proposed framework is designed to integrate supervised learning models, reinforcement learning-based optimization, and explainable AI in a relational database system to achieve SQL-native prediction with the support of cost-aware maintenance decision-making without the need to use machine learning services.

A. Dataset and Data Preparation

The study used the Predictive Maintenance dataset, which includes machine operating parameters used in predicting failures in the equipment. Some of the parameters are air temperature (T_{air}), process temperature ($T_{process}$), rotational speed (R_{speed}), torque (τ_{torque}), and tool wear (W_{tool}). The dependent variable is a measure of the machine failure.

The machine operational state can be represented as

$$\mathbf{M}_{state} = [T_{air}, T_{process}, R_{speed}, \tau_{torque}, W_{tool}] \quad (1)$$

Before the model training, data preprocessing was done to ensure the quality of the data. The irrelevant attributes, e.g., the machines, IDs, were removed through excision, whereas the categorical data, e.g., the type of machines, were transformed using one-hot encoding. The numerical data were standardized to ensure the scales of the features which were consistent.

The data was then split into the training set and the test set using the 80:20 split.

B. Supervised Model Configuration

A set of supervised algorithms was used to assess their suitability for the purpose of predicting machine failures.

The list of techniques used includes Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, AdaBoost,

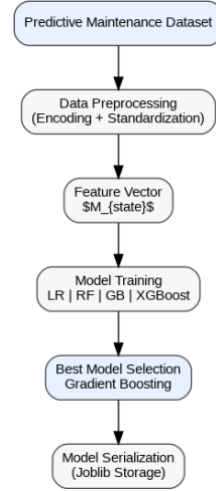


Fig. 8. Machine learning training pipeline used for predictive maintenance modelling and model selection.

XGBoost, Support Vector Machine, and Multi-Layer Perceptron.

Among the techniques, the predictive stability and non-linear relationship modeling capacity of the Gradient Boosting technique made it the most suitable choice as the primary prediction technique, which would be used in the SQL-based inference engine.

The ensemble prediction model is represented by the following equation:

$$FailureScore_m(\mathbf{M}_{state}) = FailureScore_{m-1}(\mathbf{M}_{state}) + \eta h_m(\mathbf{M}_{state}) \quad (2)$$

where $h_m(\mathbf{M}_{state})$ represents the weak learner trained on residual error and η denotes the learning rate.

The trained model was serialized using the Joblib library and stored for real-time inference.

C. SQL-Based Prediction Engine

The trained models were integrated into a database centric prediction system, thus allowing prediction analytics to be performed directly via SQL queries. In the implementation, the Streamlit library was used to provide an interface that would allow users to send SQL-based prediction queries.

When a SQL query is entered, the middleware will detect the query, thus injecting prediction functionality into the database processing pipeline. This allows the prediction operations to be carried out within the relational database system, thus avoiding the need to send the data to the prediction servers using machine learning.

The prediction functionality can be represented as

$$FailureDecision(M_{state}) = \begin{cases} 1, & \text{if } R_{speed} \leq \theta_1 \\ & \text{and } T_{torque} \geq \theta_2 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

This transformation enables Decision Tree logic to be translated into SQL CASE WHEN expressions for in-database inference.

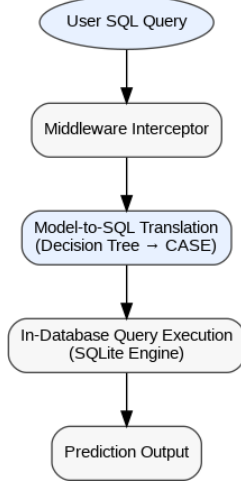


Fig. 9. Workflow of the SQL-based prediction engine enabling machine learning inference directly within the relational database system.

D. Learning-Driven Optimization

In order to support the maintenance decision-making process, the reinforcement learning module was included in the system. Here this module learns the maintenance strategies by interacting with the machine's operational state.

The machine state used by the reinforcement learning agent is defined as

$$M_{state} = [T_{air}, T_{process}, R_{speed}, \tau_{torque}, W_{tool}, P_{fail}] \quad (4)$$

where P_{fail} represents the predicted failure probability. Three reinforcement learning algorithms were implemented:

- Deep Q-Network (DQN)
- Double Deep Q-Network (Double-DQN)
- Prioritized Experience Replay DQN (PER-DQN)

The reinforcement learning agent learns an optimal policy by minimizing longterm operational cost through interaction with the environment.

$$Q_{target} = C_{reward} + \gamma \times Q_{target} (M'_{state}, \arg \max_a Q_{policy}(M'_{state}, a)) \quad (5)$$

where C_{reward} represents the maintenance cost signal and γ denotes the discount factor.

E. Model Explainability Module(XAI) Integration

To promote transparency and explainability, Explainable Artificial Intelligence (XAI) techniques were incorporated into the model. The explainability feature utilizes the SHAP method to measure the significance of the machine sensor variable to the predicted outcome. The feature importance

Algorithm	Accuracy (%)
Logistic Regression	82.50%
Decision Tree	95.95%
Random Forest	97.95%
Gradient Boosting	98.60%
AdaBoost	97.00%
Extra Trees	97.70%
XGBoost	98.85%
SVM	91.80%
MLP	98.45%
DQN	97.15%
Double-DQN	97.40%
PER-DQN	97.10%

Fig. 10. Accuracy comparison of implemented machine learning algorithms for predictive maintenance.

analysis indicates the significance of the features, including Spanning torque, rotational speed, and tool wear, up the machine failure outcome.

F. Instance-Level Explanation

Besides offering explanations at the global level, the system also offers explanations for individual machine records, i.e., at the instance level. For each prediction, the system indicates the probability of failure, as well as the explanation of the factors that affect the prediction. This helps the operators understand the prediction results.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, the proposed Intelligent SQL-ML Engine is evaluated based on the prediction accuracy of the implemented machine learning models, as well as the application of reinforcement learning for the support of maintenance decisions. In addition, the experiments will also illustrate the direct inclusion of the prediction models in the SQL query execution.

A. Performance of Supervised Learning Models

Several supervised learning algorithms were implemented using the provided predictive maintenance dataset, namely Decision Tree, Gradient Boosting, AdaBoost, Extra Trees, SVM, and MLP. The algorithms were trained using the operational variables of the machine the data considered is air temperature, process temperature, rotational speed, torque, and tool wear, and then compared to determine the best algorithms for the prediction of the machine failure.

Fig. 10 presents the comparative accuracy results obtained from the implemented algorithms.

As the above evaluation indicates, the ensemble learning models generally perform better in terms of predictability compared to the traditional linear methods. Of all the learning models, Gradient Boosting and XGBoost were the most consistent, followed by Random Forest and Extra Trees, as they are also capable of handling non-linear relationships in the sensor variables.

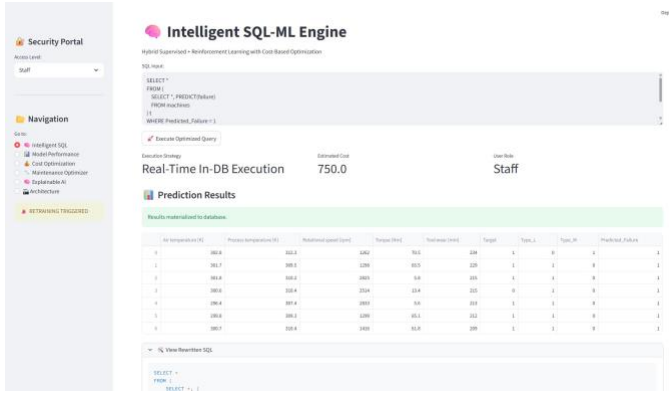


Fig. 11. Frontend interface displaying prediction results generated from SQL-based machine learning queries.

B. Reinforcement Learning Based Optimization

We have also investigated how reinforcement learning might be used to support the decision-making process of what kind of maintenance action to take. The approaches used were Deep Q-Network (DQN), Double Deep Q-Network (Double DQN), and PER DQN (Prioritized Experience Replay DQN).

The approaches learn what kind of maintenance to perform by interacting with how the machine is working and by learning the rewards, which are related to the cost and reliability of the equipment. The Double DQN approach is used to make the training more stable, and the PER DQN approach is used to make the learning faster by focusing on the most important experiences. The approaches will then be able to suggest actions such as continuing to run, preventive maintenance, or replacement of a part.

C. SQL-Based Prediction Output

An important feature of the proposed framework is the ability to carry out direct predictions using SQL queries. In the system, the parameters used in the machine operation are provided in the form of SQL queries, which are captured in the middleware layer.

The middleware layer incorporates the prediction model in the database execution process, allowing the prediction to take place in the relational database environment.

Fig. 11 illustrates the prediction output using the Intelligent SQL-ML Engine interface.

The interface shows machine sensor information and predicted operation status, which is helpful in checking the status of the machine. This interface shows how machine learning models can be used to improve the ability of SQL queries to perform predictive analytics.

VIII. FINAL SUMMARY AND FUTURE WORK

In this work, In-Database Machine Learning: A Study on SQL-Based Predictions was presented, a single framework that allows the execution of machine learning models in a relational database system using SQL. With middleware-based query rewriting of translation of Decision Tree models into SQL CASE expressions, the proposed system avoids external model

servers and can greatly decrease the data movement latency. A Cost-Based Optimizer (CBO) is used to dynamically select efficient execution strategies whereas Role-Based Access Control (RBAC) is used to provide secure and controlled access. In addition to this supervised learning, Double DQN and PER DQN were combined to enable cost optimization in the maintenance process. SHAP-based global and instance-level explanations with feature alignment in safeguarding were introduced to the concept of explainable AI (XAI). Experimental analysis on the predictive maintenance, bank marketing, and telecom churn data sets proved to be competitive in predictive accuracy and maintained in-database efficiency and maximization of cost of operation.

The future research direction will be to support distributed and cloud-native databases, make concept drift adaptation retrainable automatically and generalize SQL translation to sophisticated ensemble algorithms. The proposed architecture can also be further improved by integrating real-time streaming analytics and federated learning frameworks to scale, be more adaptive, and enterprise-ready.

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